

Cleverscope calibrating the CS448

Summary

This instruction guides you through replacing calibrating the Cleverscope CS448.

Introduction

You can either calibrate one channel at a time, or calibrate any number of channels together (1-4). The application will automatically work out which channels are being calibrated.

CS448's with serial numbers starting CR10030 onwards include a switchable voltage reference (Cal Ref) that can be used to calibrate individual channels. Cal Ref is based on a $7.5V \pm 0.035\%$ voltage reference.

Calibrating one Channel at a time using Cal Ref

Power up the CS448, plug in the USB, and get the unit capturing with the Cleverscope4 application. Allow the unit to capture at least 5 minutes, to warm up.



Using a single scope probe, set to 1x, use Settings/Calibrate Cleverscope, and choose the Single Channel tab.

Find the Cal Ref and 0V holes on the left hand side of the CS448 front panel.

Take the spring ground clip out of the probe accessory bag and twist it onto the probe as shown.

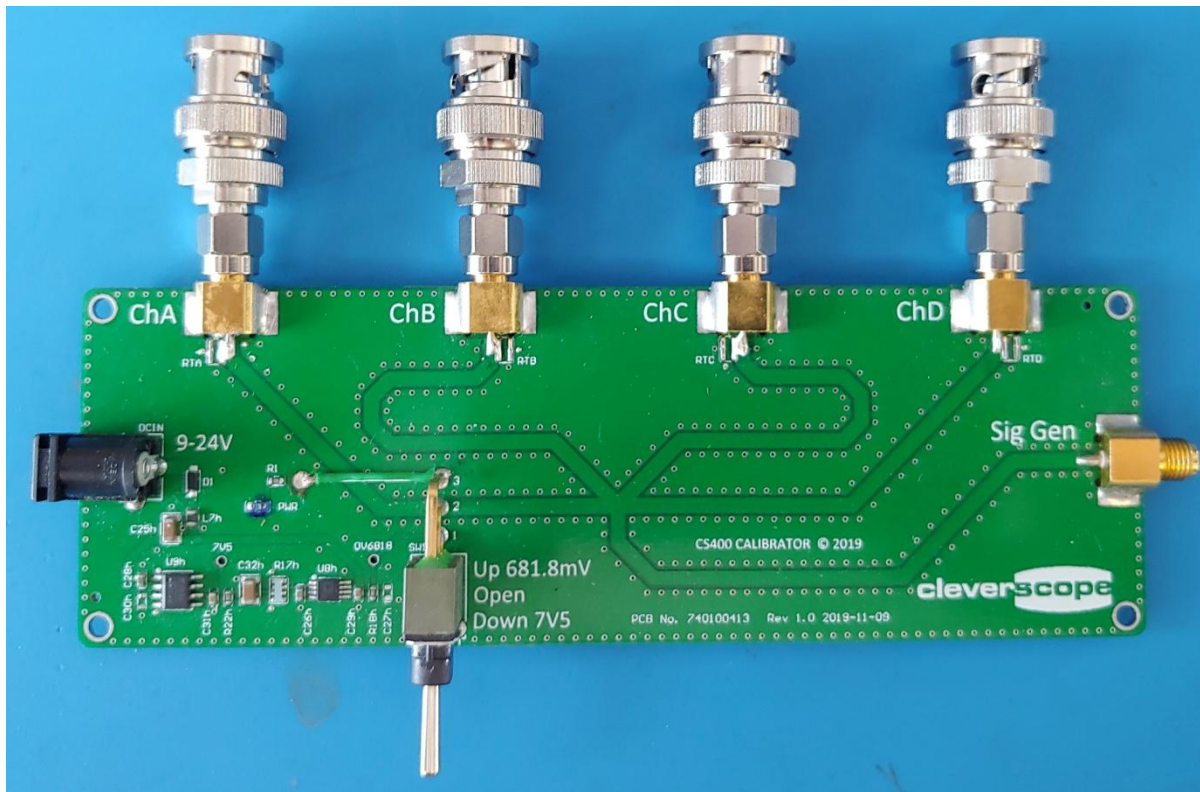
Connect the Probe to Channel A. Insert the probe tip into the Cal Ref hole, with the spring ground going into the 0V hole. Both need to be connected. Click on 'Calibrate'. After a short time the Cal Complete will turn on green.

Next complete this process for Channels B - D.

If there is an error, the Cal Error led will light red. Makes sure both the probe tip and spring probe are touching the surface inside the holes. You do not need much force. Try again.

Calibration of 1-4 channels using the Cleverscope Calibrator

You can use the Cleverscope Calibrator to calibrate all four channels of the CS448 together. You will need to obtain a Calibrator from Cleverscope.

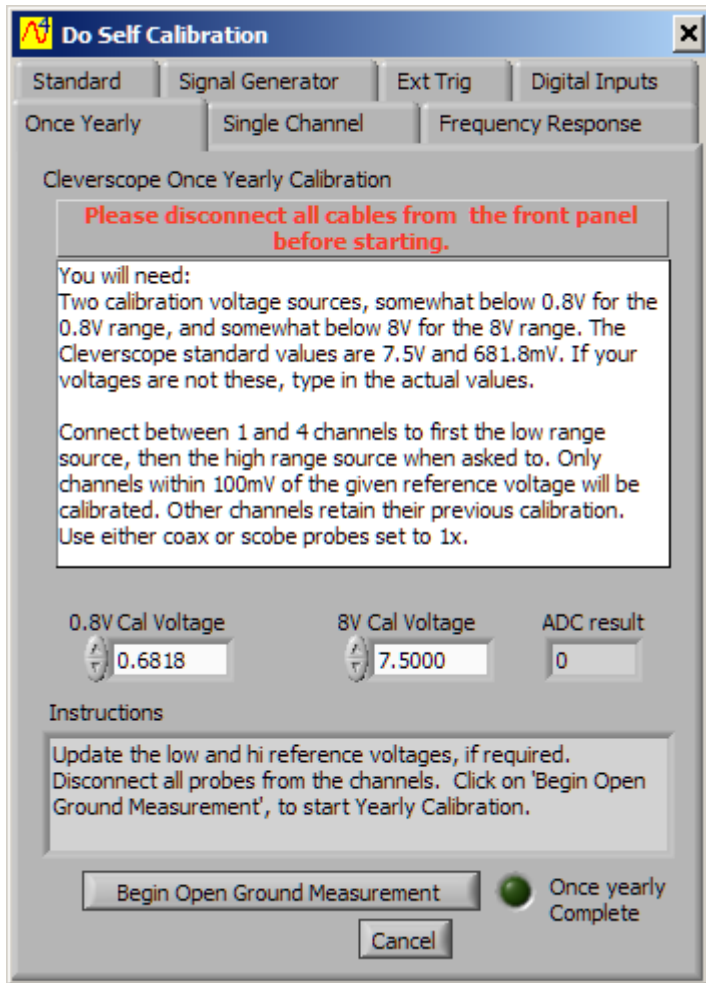


The calibrator has a 9-24V DC input to power the reference. You can use a Cleverscope 9, 12 or 19V power supply for this (centre pin positive). The three way switch chooses between Open (center position), 681.8mV (switch up) or 7.5V (switch down). There is also a sig gen connection for calibrating the signal generator, or for checking channel time alignment. The calibrator uses the same $7.5V \pm 0.035\%$ voltage reference.

Plug the Calibrator into the CS448, and attach a power supply to it. Allow it to warm for 5 minutes:



Run the scope at least 5 minutes to warm up. Choose Settings/Calibrate Cleverscope, and choose the Once Yearly Tab:



Carry out these steps:

1. Have the switch in the middle, open position, and click the 'Begin Open Ground Measurement' Button.
2. Once the Button name changes to 'Begin 0.8V Range Cal Measurement'; flip the switch up to the 0.6818V position. Click the Button.
3. Once the Button name changes to 'Begin 8V Range Cal Measurement', change the switch down to the 7.5V position. Click the Button. On completion the 'Once yearly Complete' led will light. Calibration is complete.
4. Click Finish to exit. Remove the Calibrator. The CS448 is ready to use.

Calibration without the Cleverscope Calibrator.

Follow the same sequence as using the Cleverscope Calibrator, but use an external reference instead – you will need one about 0.68V and another about 7.5V. Enter the exact voltages into the Do Self Calibration panel.

Follow the same sequence, but apply open, 0.6818 and 7.5V to the input, or inputs as you click the Button to change to the next voltage.

Calibrating the Signal Generator

To Calibrate the signal generator, connector a 1x probe, or Coax from the signal generator output (usually the filtered output) to Channel A. Click the Signal Generator tab on the Do Self Calibration panel. Follow the instructions. You can also connect the Signal Generator to the right hand socket on the Cleverscope Calibrator:



Make sure you leave the switch in the centre position. It does not matter if the Calibrator is powered.